For a **500 MW/year polycrystalline silicon solar cell production line** (wafer → solar cell, excluding polysilicon and wafer manufacturing), the total investment depends strongly on technology level, automation, and whether you buy Chinese turnkey equipment or European/Japanese high-end equipment. Current industry estimates put a fully imported 500 MW cell line in the range of **US$25–45 million for equipment only** and around **US$40–70 million including factory infrastructure and commissioning**. (Scribd)

Below is a realistic 2026 estimate.

1. Production Scope Assumption

Capacity: 500 MW/year
Technology: Polycrystalline silicon solar cell (traditional Al-BSF / PERC type)
Input: Silicon wafers
Output: Finished solar cells
Working days: 330 days/year
Efficiency target: 18–20% (polycrystalline)

Approximate output:

- Cell size: 156–166 mm wafer
- Annual cells: ~25–30 million pieces
- Daily production: ~80,000–100,000 cells

A. Main Production Equipment Cost

Process	Equipment	Qty	Estimated Cost
Wafer loading	Automatic wafer loader	2–3	$0.5–1M
Cleaning	RCA cleaning / wet bench	3–5	$2–4M
Texturing	Acid/alkaline texturing line	1	$2–3M
Diffusion	POCl3 diffusion furnace	4–6	$5–8M
PSG removal	Wet etching system	1	$1–2M
Edge isolation	Laser edge isolation	2–3	$1–2M
PECVD	SiNx anti-reflection coating	3–5	$8–12M

Process	Equipment	Qty	Estimated Cost
Screen printing	Ag front + Al back printing	3–5	$6–10M
Firing furnace	Belt firing system	3–5	$3–5M
Sorting	IV tester + EL tester	2–4	$2–4M
Automation	Conveyors, robots, MES	-	$3–6M

Equipment subtotal:

≈ US$35–55 million

Chinese suppliers can be cheaper; European/Japanese suppliers can exceed this range. (OECD)

B. Factory Infrastructure

Building

500 MW cell factory requires:

- Cleanroom area: 8,000–15,000 m²
- Total building: 15,000–25,000 m²

Cost:

Item	Cost
Factory building	$8–15M
Cleanroom HVAC	$5–10M
Chemical room	$2–4M
Gas system	$2–5M
Wastewater treatment	$2–5M
Electrical substation	$3–6M

Infrastructure:

≈ $25–40 million

C. Engineering and Commissioning

Item	Cost
Installation	$3–5M
Training	$1–2M
Process optimization	$2–4M
Spare parts	$2–3M

Subtotal:

≈ $8–14 million

D. Working Capital

Initial inventory:

- Silicon wafers
- Silver paste
- Aluminum paste
- Chemicals
- Gases
- Packaging

Typical:

$10–20 million

Total Investment

Option 1 — Chinese Turnkey Line

Item	Cost
Equipment	$30–40M
Factory	$20–30M
Installation	$5–8M
Working capital	$10M

Total:

≈ US$65–90 million

Option 2 — Premium International Line

Item	Cost
Equipment	$45–60M
Factory	$30–45M
Installation	$10M
Working capital	$15–20M

Total:

≈ US$100–135 million

If You Want a Fully Integrated Plant

If you also produce:

1. Polysilicon

Additional:
$150–300M

2. Silicon ingot growth

Additional:
$30–60M

3. Wafer slicing

Additional:
$40–80M

A complete:

Quartz → polysilicon → ingot → wafer → cell → module

500 MW/year factory:

≈ US$300–500 million

Operating Cost Estimate (500 MW/year)

Annual production:

500 MW

Approximate cost:

Item	$/W
Silicon wafer	0.08–0.12
Silver paste	0.015–0.025
Chemicals/gases	0.005–0.01
Electricity	0.01–0.02

Item	$/W
Labor	0.01–0.02
Maintenance	0.01

Cell manufacturing cost:

≈ $0.13–0.20/W

Annual production cost:

500,000,000 W × $0.15/W

≈ **$75 million/year**

My estimate for a new factory in China/Asia:

For a competitive 500 MW polycrystalline solar cell plant:

CAPEX:
 Equipment: **$35–45M**
 Factory + utilities: **$25–35M**
 Total project: **$70–90M**

This is the realistic range for a modern Chinese turnkey project. (Alibaba)

I can also provide a **complete machine list with manufacturer names (Jinggong, RENA, Centrotherm, Maxwell, Schmid, etc.) and individual machine prices for a 500 MW/year line** if you need a feasibility study.